Questions

Q1.

A diver is swimming underwater in a lake.

The diver wears the meter shown in Figure 3.

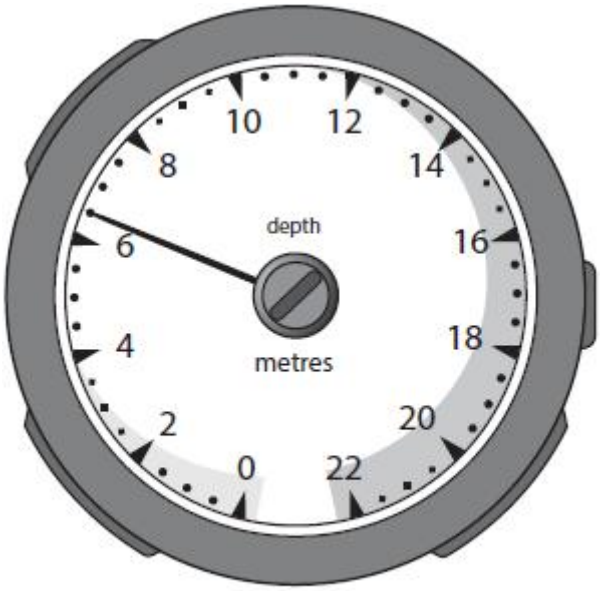

Figure 3

(i) The meter shows the depth of the diver below the surface of the water.

State the depth shown on the meter in Figure 3.
State the unit.

(2)

depth = ...unit = ...

(ii) State how the pressure of the water on the diver changes as the diver swims down from the surface of the
lake.

(1)

.....

.....

(iii) State why the total pressure on the diver is greater than just the pressure due to the water above the diver.

(1)

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(Total for question = 4 marks)

Q2.

Figure 12 shows what happens when weights are added to the end of a wooden strip.

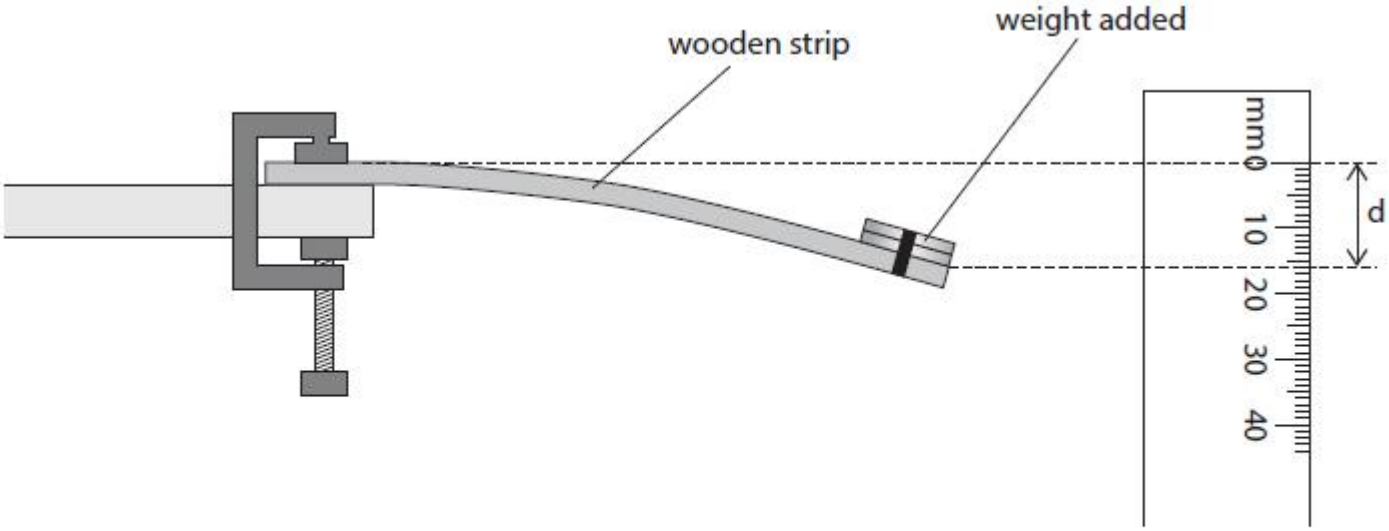

Figure 12

(Source: https://www.schoolphysics.co.uk/age16-19/Mechanics/Simple%20harmonic%20motion/text/Vibrating_cantilever/index.html)

The end of the strip of wood moves down by a distance, d, as shown in Figure 12.

(i) Describe how a student could investigate how the distance, d, depends on the weight added to the wooden strip.

(4)

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(ii) A student conducts a second experiment with the same equipment where they decide to keep the size of the weight constant.

State another variable, other than the weights added, that would affect the distance, d.

(1)

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(Total for question = 5 marks)

Q3.

The graph in Figure 14 shows the variation in atmospheric pressure with the height above sea level.

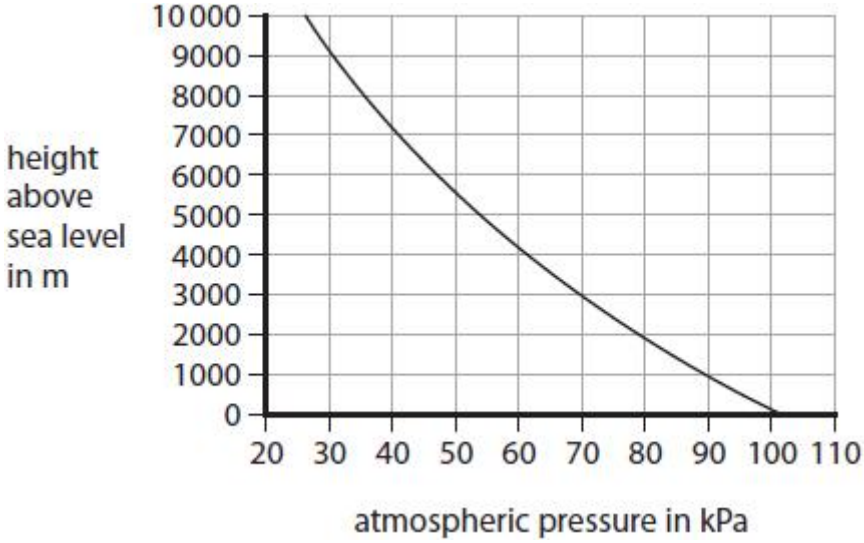

Figure 14

(i) Use the graph to estimate the atmospheric pressure at 3000 m above sea level.

(1)

atmospheric pressure = ... kPa

(ii) Use the graph to estimate the atmospheric pressure at 6000 m above sea level.

(1)

atmospheric pressure = ... kPa

(iii) Suggest a reason why the atmospheric pressure decreases with height above sea level.

(1)

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.....

(Total for question = 3 marks)

Q4.

This question is about pressure.

Figure 22 shows windows in an aeroplane.

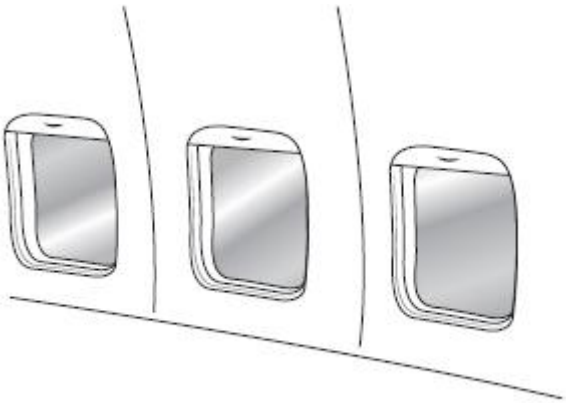

Figure 22

The aeroplane is high above the Earth's surface.

The atmospheric pressure outside the aeroplane is 23 000 Pa.

The air pressure inside the aeroplane is 80 000 Pa.

(i) Calculate the pressure difference between inside and outside of the aeroplane.

(1)

pressure difference = ... Pa

(ii) The surface area of the window is 0.094 m^2.

Calculate the size of the force on the window due to the cabin air pressure of 80 000 Pa.
Use the equation

$$P = \frac{F}{A}$$

(2)

force = ... N

(iii) On the same aeroplane, a different window has a smaller surface area.

Explain how the force due to the air pressure inside the cabin on the small window differs from the force on the larger window.

(2)

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(iv) Figure 23 shows a cross-section through the aeroplane including one window.

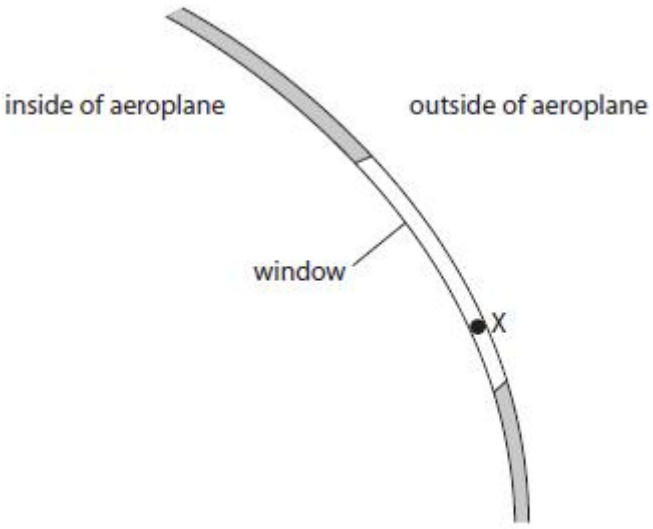

Figure 23

Draw an arrow on Figure 23 to show the direction of the resultant force due to the air pressure inside the cabin on the window at point X.

(2)

(Total for question = 7 marks)

Q5.

A box weighs 300 N.

The box rests on the floor.

The area of the box in contact with the floor is 0.75 m^2.

(i) Calculate the pressure exerted by the box on the floor.

Use the equation

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

(2)

pressure = ... Pa

(ii) Which of these is a unit of pressure?

(1)

- ☐ **A** kg/m^2
- ☐ **B** $kg\ m^2$
- ☐ **C** N/m^2
- ☐ **D** $N\ m^2$

(Total for question = 3 marks)

Q6.

Figure 23 shows some water in a tank.

Figure 23

(i) The bottom of the tank has an area of 0.80 m^2.

The force on the bottom of the tank, due to the water, is 2400 N.
Calculate the pressure, due to the water, on the bottom of the tank.

(3)

pressure = ... Pa

(ii) More water is added to the tank.

Explain how the pressure on the bottom of the tank changes when more water is added to the tank.

(2)

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.....

(iii) Figure 24 shows an object on the bottom of the tank of water.

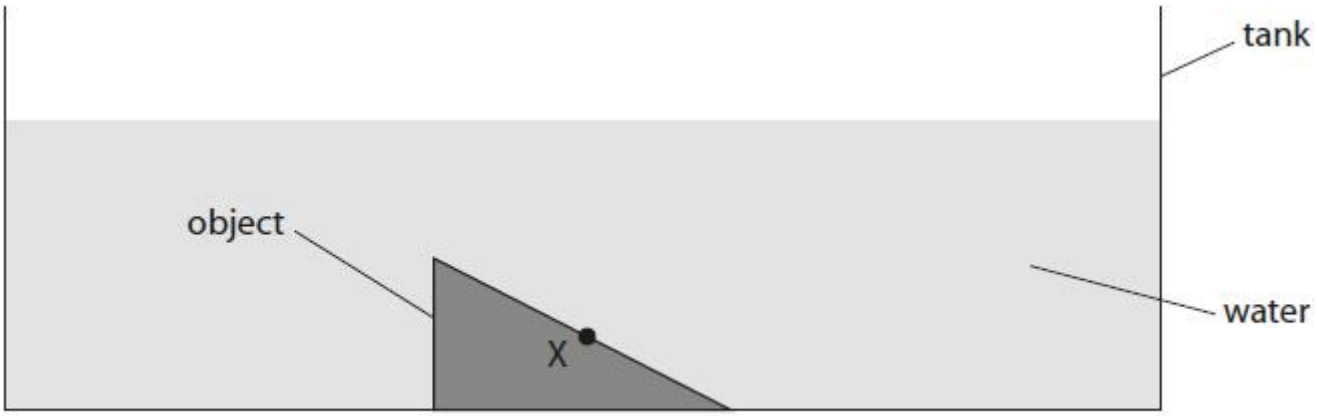


Figure 24

Draw an arrow on Figure 24 to show the direction of the force exerted by the water on the surface of the object at point X.

(1)

(Total for question = 6 marks)

Q7.

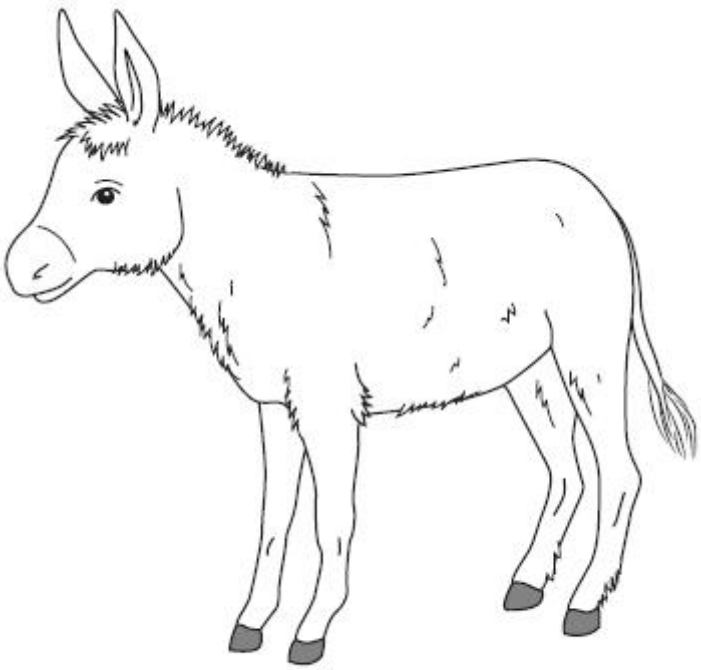


Figure 23

A donkey has a weight of 2500 N.

The area of each hoof is 0.022 m^2.

(i) Calculate the average pressure that the donkey exerts on the ground.

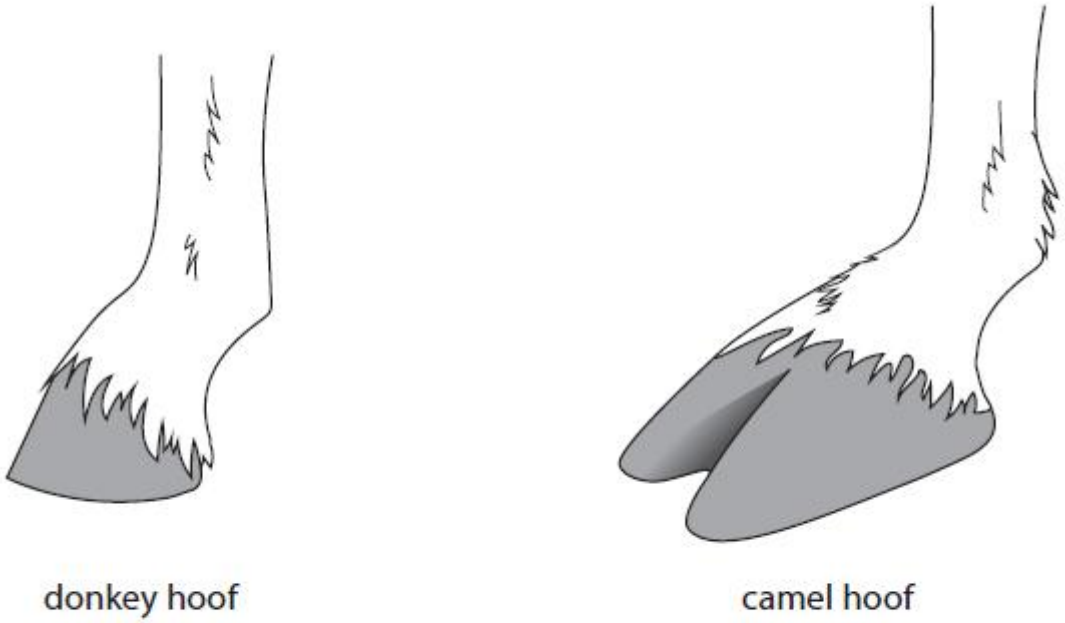
Use the equation

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

(2)

average pressure = ... Pa

(ii) Figure 24 shows how the shape of a camel's hoof is different from the shape of a donkey's hoof.



Both diagrams are drawn to the same scale.

Figure 24

The camel and the donkey have the same mass.

Explain how a camel's hoof is a more suitable shape than a donkey's hoof for walking on soft ground.

(2)

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(Total for question = 4 marks)

Q8.

Figure 26 shows an ice skater standing on one skate.

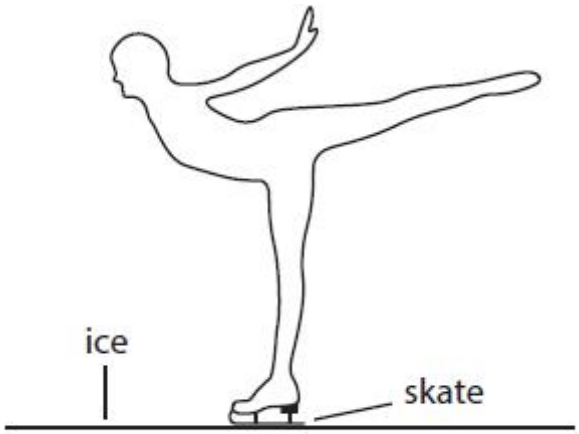


Figure 26

Calculate the force the skate exerts on the ice.

pressure of skate on ice = 4.8×10^7 Pa

area of blade in contact with ice = 1.2×10^{-5} m^2

Use the equation

force = pressure × area

Give your answer to 2 significant figures.

(3)

force = ... N

(Total for question = 3 marks)

Q9.

Figure 17 shows a bicycle pump with a closed end.

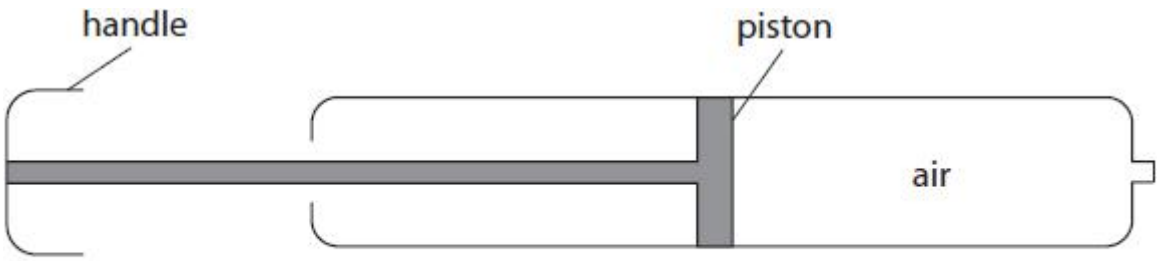


Figure 17

The area of the piston is 2 cm^2.

A force of 28 N is applied to the piston.

Calculate the pressure, in N/cm^2, of the piston on the air in the bicycle pump.

(3)

pressure = ... N/cm^2

(Total for question = 3 marks)

Q10.

Figure 4 shows a container of length 6.0 m and width 2.0 m resting on a floor.
The weight of the container is 15 000 N.

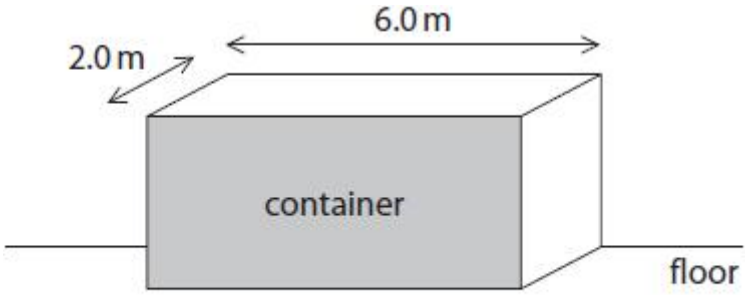


Figure 4

Calculate the pressure that the container exerts on the floor.

Use the equation

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

(3)

pressure of the container on the floor = ... Pa

(Total for question = 3 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
i	6.5 (1) m (1)	 metres / meters allow M independent marks	(2)

Question number	Answer	Additional guidance	Mark
ii	the pressure increases		(1)

Question number	Answer	Additional guidance	Mark
iii	there is additional pressure due to the atmosphere		(1)

Q2.

Question Number	Answer	Additional guidance	Mark
(i)	a description including any **four** of the following: measure **initial** d / distance / height (1) change weight (1) measure **new** d / distance / height (1) repeat / continue (changing weight and measuring d) (1) plot a graph (d against weight) (1)	 accept reference to 16 (mm)	(4) AO3a AO 3.2a/2b

Question Number	Answer	Additional guidance	Mark
(ii)	any **one** from: distance between the clamp and the masses (1) (change) strip (1)	accept ruler for strip throughout accept length of the strip accept thicker strip /thinner strip / material of strip / strip of different density etc.	**(1)** **AO3.3b**

Q3.

Question Number:	Answer	Mark
(i)	70 (kPa)	**(2)** AO 3 1a

Question Number:	Answer	Mark
(ii)	between 46 and 48 (kPa)	**(1)** AO 3 1a

Question Number:	Answer	Additional guidance	Mark
(iii)	any one from: the atmosphere gets less dense / thinner / has fewer molecules (as height above sea level increases) (1) or there is less air/oxygen (as the height above sea level increases)(1)	 accept particles less weight pushing down	**(1)** AO 1 1

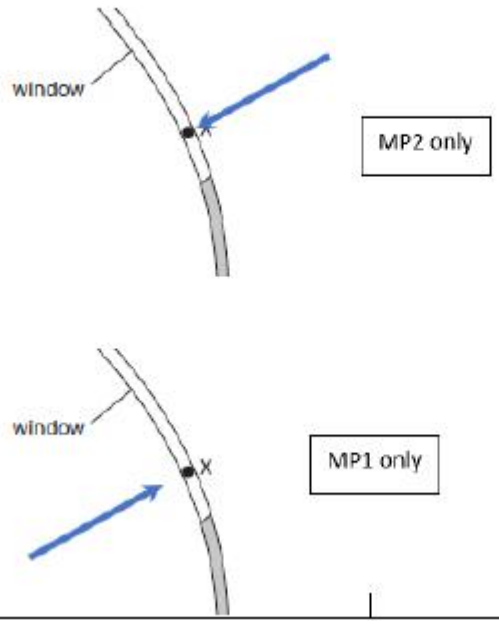
Q4.

Question number	Answer	Additional guidance	Mark
(i)	(80 000 – 23 000) 57 000 (Pa) (1)	 -57 000 (Pa)	**(1)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) $80\ 000 = \frac{F}{0.094}$ rearrangement and evaluation (1) (F=) 7500 (N)	alternative method re-arrangement (1) (F =) P x A **or** (F=) 80 000 x 0.094 (substitution and) evaluation (1) accept 7520 (N), award full marks for correct answer without working. allow 1 mark max for substitution using pressure of 57 000 **or** an answer that rounds to 5400 e.g. 5358 (calculated net force)	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(iii)	force is less (on small window) (1) pressure is the same (1)	force is greater on large window	**(2)** **AO1**

Question number	Answer	Additional guidance	Mark
(iv)	arrow pointing towards outside of aeroplane (1) arrow is normal to surface at X (judge by eye) (1)	may be inside or outside of aeroplane. need not touch X do not award if two or more conflicting arrows drawn must touch X or dot at X independent marks	**(2)** **AO1**

	Examples:  		

Q5.

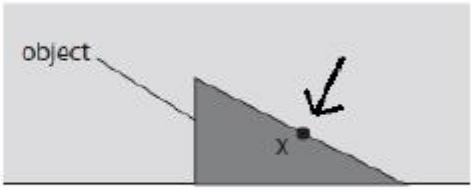
Question number	Answer	Additional guidance	Mark
(i)	substitution (1) $\frac{300}{0.75}$ evaluation (1) 400 (Pa)	Award full marks for correct answer without working	(2)

Question number	Answer	Mark
(ii)	**D** N/m^2	(1)

Q6.

Question number	Answer	Additional guidance	Mark
i	recall (1) $p = \frac{F}{A}$ substitution (1) $(p) = \frac{2400}{0.8}$ evaluation (p) = 3000 (Pa) (1)	may be implied by a correct substitution award full marks for the correct answer without working	**(3)** **AO2**

Question number	Answer	Additional guidance	Mark
ii	an explanation linking greater pressure (on bottom of tank) (1) with greater force **due to water** (above bottom of tank) (1)	 more weight of water more depth/height of water ignore simply 'more water' or 'greater amount of water'	**(2)** **AO1**

Question number	Answer	Additional guidance	Mark
iii	 an arrow perpendicular to the sloping side **and** pointing towards X	 judge by eye	**(1)** **AO1**

Q7.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) (pressure =) $\frac{2500}{4 \times 0.022}$ evaluation (1) 28 000 (Pa)	 any number rounding to 28 000 e.g. 28 400, 28 410, 28 409 award full marks for the correct answer without working award **one** mark for numbers that round to 110 000 (Pa) (missing 4 in denominator) award 1 mark for 454 545 (times by 4)	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	An explanation linking any **two** from camel less likely to sink into the soft ground (1) (same) force / weight is distributed / spread out (1) camel's hoof has greater (surface) area (than donkey) (1) camel's hoof exerts less pressure (than it would if hoof were smaller) (1)	ORA for donkey ignore pressure is spread out wider if no other marks scored then allow 1 mark for split in camel hoof enables better grip (as it walks)	**(2)** **AO3**

Q8.

Question	Answer	Additional guidance	Mark
	substitution (1) (force =) 4.8 (x 10^7) x 1.2 (x 10^{-5}) evaluation (1) 576 (N) their evaluation rounded to 2sf (1) 580 (N)	award full marks for the correct answer (580) without working award 1 mark for 5.76 to any other power of ten award 2 marks for 5.8 to any other power of ten	**(3)** **AO2.1**

Q9.

Question number	Answer	Additional guidance	Mark
	recall equation (1) pressure = $\frac{\text{force}}{\text{area}}$ substitution (1) (pressure =) $\frac{28}{2}$ evaluation (1) 14 (N/cm^2)	 accept $P = \frac{F}{A}$ ignore powers of ten errors for attempted unit conversions award full marks for the correct answer without working if no other marks scored then allow 1 mark for either $\frac{2}{28}$ or 0.07	**(3)**

Q10.

Question Number	Answer	Additional guidance	Mark
	(area) = 6.0 x 2.0 = 12 (1) substitution (1) (P=) $\frac{15\,000}{(12)}$ evaluation (1) 1300(Pa)	award one mark for 6.0 x 2.0 seen with no alternative area calculation Accept 15000/(any value) for this mark. accept 1250 (Pa) award full marks for the correct answer without working	**(3)**